KAPASA MAKASA UNIVERSITY

CYBER SECURITY

ACCESS CONTROL & INTRUSION DETECTION (CYS335)

EXPECTED QUESTIONS

Questions by James Soko

CHAPTER ONE

UNIT 1 – INTRUSION DETECTION SYSTEM (IDS)

1. Briefly highlight the history of intrusion detection systems. [4 Marks]
2. What do you understand by the term intrusion detection system (IDS)? [2 Marks]
3. What is manifestation in relation to intrusion detection systems (IDS)? [2 Marks]
4. List and explain the four (4) detection techniques by highlighting their advantages and disadvantages. [8 Marks]
5. An intrusion detection system's response is its output or action upon detecting a problem. A response can take many different forms, list and explain the two forms. [2 Marks]
6. List and explain the two types of threats. [2 Marks]
7. List and explain the types of intrusion detection systems. [5 Marks]

TOTAL: [25 Marks]

8. Give at least three (3) needs of an intrusion detection system. [3 Marks]
9. List and write few notes on any three (3) top intrusion detection tools. [3 Marks]

UNIT 2 – INTRUSION PREVENSION SYSTEM (IPS)

1. What is an intrusion prevention system? [2 Marks]
2. How do intrusion prevention systems work? [3 Marks]
3. List and explain the two (2) methods used by intrusion prevention systems to detect malicious activities. [2 Marks]
4. List and explain the three (3) types of prevention techniques. [3 Marks]
5. List and explain the types of intrusion prevention systems. [4 Marks]

TOTAL: [20 Marks]

6. Explain the anatomy of intrusion analysis. [4 Marks]
7. What is vulnerability assessment? [2 Marks]
8. List and explain three (3) importance of vulnerability assessment. [3 Marks]
9. List and explain at least three (3) types of vulnerability assessments. [3 Marks]
10. Differentiate between credential and non-credential analysis. [2 Marks]

TOTAL: [14 Marks]

UNIT 3 – 5 – SNORT

Questions 1 (Unit 3)

1. What is snort? [2 Marks]
2. Who developed snort? [1 Mark]
3. In which year was snort developed? [1 Mark]
4. Which networking company now develops snort? [1 Mark]
5. List and explain at least four (4) snort features. [4 Marks]
6. List and explain at least ten (10) snort commonly used command line options. [10 Marks]
7. List and explain at least six (6) snort alert modes. [6 Marks]

TOTAL: [25 Marks]

Question 2 (Unit 4)

1. Draw the structure of a snort rule and explain each and every component. [7 Marks]
2. List and explain at least five (5) actions in a snort rule. [5 Marks]
3. Highlight at least three (3) snort rule snippets that can use an IP address 172.24.20.0/20, $HOME_NET, $EXTERNAL_NET, 'any' or port number e.g., 21 in its syntax. [5 Marks]
4. What is a snort configuration file? List and explain the six (6) basic components of this configuration file. [8 Marks]

TOTAL: [25 Marks]

Question 3 (Unit 5)

Questions will be updated once the summary is available.

Questions 4 (Practical Lab)

1. You are the IT System Manager for Kapasa Makasa University, and recently they have been struggling with cyberattacks. With your experience in threat detection and prevention, you are tasked to –
    a) Identify your active network interface. [1 Mark]
    b) Run Snort in packet sniffing mode using the configuration file by showing all necessary commands and configurations (hint: using steps e.g., Step 1). [5 Marks]
    c) Write a command that will generate some traffic toward the Snort machine by performing a ping or a network scan. [1 Mark]
    d) After realizing that most of the snorts default rules are not effective for the attacks targeting your network. You are tasked to also configure –
        i. ICMP local rule that will detect the ping or ARP spoofs on your network. [2 Marks]
        ii. Show how you can make the ICMP local rule active by configuring the snort. [3 Marks]
        iii. At this point your network hold critical information and you don't just want to detect but also prevent unauthorized access. Provide a sample Snort rule to detect HTTP traffic on port 80 and explain its structure. [5 Marks]
        iv. Run Snort in inline mode (ensure interface and bridge setup if required) [3 Marks]

    TOTAL: [20 Marks]

2. What do the fields sid and rev represent in a Snort rule? [2 Marks]
3. What effect does changing a Snort rule from alert to drop have on network traffic? [2 Marks]
4. Explain how Snort detects and classifies suspicious packets. [3 Marks]
5. Describe a scenario where you would use Snort in IPS mode instead of IDS mode. [3 Marks]

TOTAL: [10 Marks]

CHAPTER TWO

UNIT 6 – ACCESS CONTROL

1. What is access control? [2 Marks]
2. Define the following terms.
    a. Identification [1 Mark]
    b. Authentication [1 Mark]
    c. Authorization [1 Mark]
    d. Access [1 Mark]
    e. Object [1 Mark]
    f. Subject [1 Mark]
    g. Operation [1 Mark]
    h. Owner [1 Mark]
    i. Custodian [1 Mark]
    j. End-user [1 Mark]
3. What is Mandatory Access Control (MAC)? [2 Marks]
4. List and explain two elements of MAC. [2 Marks]
5. How can you determine if a file can be opened in MAC? [2 Marks]
6. List two implementations of MAC. [2 Marks]

TOTAL: [20 Marks]

7. What is Discretionary Access Control (DAC)? [2 Marks]
8. What is Non-discretionary Access Control? [2 Marks]
9. What's the main difference between Discretionary and Non-discretionary Access Control? [2 Marks]
10. What is Rule Based Access Control (RBAC)? [2 Marks]
11. Give at least four best practices of access control. [4 Marks]
12. What is an Access Control List (ACL)? [2 Marks]
13. What do we call each entry in an ACL table? [1 Mark]

TOTAL: [15 Marks]

14. Authentication services that provide Authentication Authorization and Accounting (AAA) include Remote Authentication Dial-In User Service (RADIUS), Terminal Access Controller Access Control System Plus (TACACS+), Kerberos and Lightweight Directory Access Protocol (LDAP).
    a) Write few notes on RADIUS by illustrating how it operates. [5 Marks]
    b) In a table format, highlight at least five (5) differences between RADIUS and TACACAS+. [5 Marks]
    c) Explain how Kerberos works by using a diagram. [5 Marks]
    d) Write few notes on LDAP. [5 Marks]

TOTAL: [20 Marks]

Questions are not complete, they will be updated once the summary is done.

Warning:

You are advised to study the lecture slides and notes, I might fail to summarize the notes before the exam, all the best!!!